# JUYEONG LEE

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## SUMMARY

AI engineer and quant researcher: multi-factor alpha, hedging and allocation models at QRAFT; at EY leads retail AI products from client requirements to daily use. Palantir Foundry & AIP certified; first author on DiffGerber, co-author on five more ICAIF 2026 papers.

## EXPERIENCE

**EY Consulting (Ernst & Young)** — Seoul, South Korea

*AI Consultant (Forward Deployed Engineer), AI & Data, Technology Consulting* — Jan 2026 – Present

*Andar (Korean athleisure brand): zero-to-one SCM decision system for forecasting, inventory and replenishment (Databricks, Neo4j)*

- **Requirements to roadmap** — Interviewed planners and MDs, turned their daily questions into the product scope and build order (ontology, agents, then forecasting) and a phased roadmap (lookup, drill-down, simulation); aligned executives and planners on what each release could answer and on the projected full-rollout impact: operating cost down 15–20%, on-time in-full from 64% to 80%.
- **SCM ontology, agents & forecasting** — Leading the EY build: a Neo4j ontology of SKUs, seasons, promotions, reorder grades and days of inventory; a supervisor agent routing planners' questions to specialist agents (SCM monitoring, cost breakdown, material and product demand forecasting); and the product-level demand forecasting model (in-season/off-season). Three Genie agents and the supervisor are in daily use; a 30-question golden set gates each release (answer accuracy up from 50% to 90%).

*Lotte Mart (Korean hypermarket chain): one-to-hundred scale-out of a production demand-forecasting system (in progress)*

- **Forecasting scale-out** — Leading the AI build that extends the weekly store × SKU forecasting pipeline (BigQuery) from hypermarkets to Lotte Super and feeds ordering and inventory decisions; managing executive and planner expectations on accuracy and rollout.

*Samsung C&T Fashion (PoC): led and demoed a Neo4j merchandising ontology and GraphRAG agent for multi-step questions*

**QRAFT Technologies (AI Hedge Fund)** — Seoul, South Korea

*Researcher (AI / Quant), Investment Product Management* — Nov 2021 – May 2025

*Multi Asset Team* — Jul 2024 – May 2025

- **Client regime signals** — Built ML models that flag risk-on/off regimes from market, macro and stock-level data and delivered client-specific equity/cash allocation signals to Samsung Securities, KB Securities, Hana Life and Simplify Asset Management (US).
- **Strategy clustering** — Grouped quantitative investment strategies (QIS) with PMFG graph-based hierarchical clustering to remove overlapping exposures and recombine them as regimes shift; validated in proofs of concept with global financial institutions.

*Long-Term Equity Team* — Nov 2021 – Jun 2024

- **SoftBank US equity long/short alpha pipeline** — Built the fund's statistical multi-factor algorithm by reducing 360 candidate strategies on financial-ratio, profitability and value factors to an orthogonal set: VIF, multiple regression and DBHT clustering removed collinear and overlapping factors before t-statistic and A/B screening; the fund delivered excess return over SPY (figures confidential).
- **Factor and risk model replication** — Reimplemented Fama-French (97% match) and the Barra risk model from scratch.
- **FSITC (Taiwanese asset manager) hedging framework** — Led the hedging-signal framework for its $200M Japan fund: an ML, GNN and LSTM-autoencoder ensemble over equity, option-implied and market volatility; in a 2006–2024 TOPIX backtest it cut maximum drawdown from −62% to −33% and volatility from 21% to 11%.
- **Multi-agent risk reporting** — Automated FSITC's weekly risk report on top of the hedging signals with three LLM agents on QRAFT's in-house harness: signal interpretation, market commentary and a risk manager that combines both into the risk assessment; also scored real-time news with chain-of-thought and few-shot prompting for a long/short strategy.
- **Trading simulator** — Built the backtesting simulator that models liquidity, slippage and market impact, raising agreement with realized P&L to 93%; used to backtest the SoftBank fund's strategies.

**UNIST Data Analytics Lab | NAVER Maps Joint Research Project** — Ulsan, South Korea

*AI Researcher (MSc)* — Sep 2019 – Sep 2021

- **Traffic congestion prediction** — Led the joint research with NAVER Maps: built a time-series pattern-matching model combined with Graph Neural Networks (GCN/GAT) that predicts congestion after traffic incidents at 82% accuracy, 15% above the benchmark, and used LIME to explain the drivers; published as first author in Transportmetrica B.

## RESEARCH & PUBLICATIONS

- **ICAIF 2026** (ACM International Conference on AI in Finance; accepted, oral), first author: DiffGerber — decision-focused learning of the Gerber covariance threshold from realized portfolio risk; Sharpe 0.44 to 0.54 on the top-100 US stocks
- **Co-authored papers** (ICAIF 2026, accepted) — AlphaLeak (oral; name-masked "blind" LLM trading benchmarks are re-identifiable from same-date news); CallRank (oral; language-model embeddings of earnings-call Q&A for sector rotation); SlipClock (displayed-depth execution-cost benchmark for crypto-futures AI strategies); The Division of Research (information access as a design variable for LLM investment teams); Metropolitan Housing Signals for Treasury Duration Risk Management.
- **Journals** (first author): *Quantifying incident impacts and identifying influential features in urban traffic networks*, Transportmetrica B, 2022; *Prediction of problematic smartphone use: A machine learning approach*, IJERPH, 2021
- **Patents** (granted in Korea and Japan): *Predicting road congestion propagation using pattern matching*, lead inventor, KR 10-2608343, JP 7453296; *Predicting propagation time delay lag of road congestion using transfer entropy*, co-inventor, KR 10-2604575, JP 7453297

## EDUCATION

**Ulsan National Institute of Science and Technology (UNIST)** — *MSc, Industrial Engineering* — Sep 2019 – Sep 2021

**Handong Global University** — *BA, Management and Economics (double major)* — Mar 2014 – Feb 2019

## SKILLS & CERTIFICATIONS

**Data & ontology:** Neo4j / Cypher, Databricks (Unity Catalog, Genie, Agent Bricks, Lakeview, Jobs), SQL / Spark SQL, Airflow, GCP
**AI / ML:** Python, PyTorch, TensorFlow, scikit-learn, pandas, LangChain, LangGraph, LLM agents, GNNs, time-series modelling
**Certifications & languages:** Palantir Foundry & AIP (Aug 2025); Korean (native); English (fluent; parliamentary interpreter, 2019)